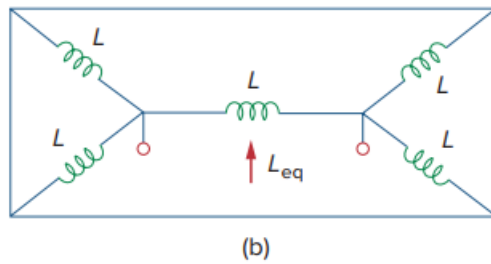
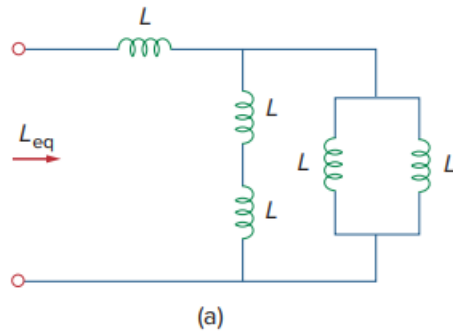


Problem

Find L_{eq} in each of the circuits in Fig. 6.77.

Fig. 6.77:



Step-by-step solution

Step 1 of 7

(a)

Refer to Figure 6.77(a) in the textbook.

In the Figure 6.77(a), the two inductors are connected in parallel. The equivalent inductance is,

$$\begin{aligned} L_{eq1} &= (L) \parallel (L) \\ &= \frac{L \times L}{L + L} \\ &= \frac{L^2}{2L} \\ &= 0.5L \end{aligned}$$

In the Figure 6.77(a), the two inductors are connected in series. The equivalent inductance is,

$$\begin{aligned} L_{eq2} &= L + L \\ &= 2L \end{aligned}$$

Step 2 of 7

Draw the reduced circuit of Figure 6.77(a).

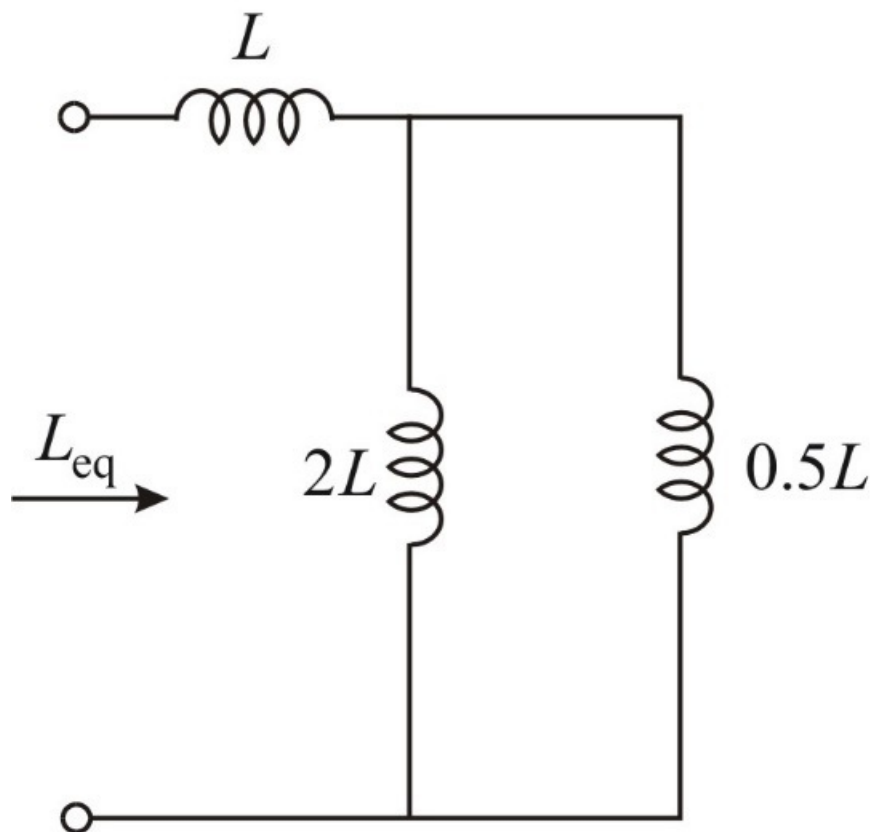


Figure 1

Step 3 of 7

In Figure 1, the parallel combination of $2L$ and $0.5L$ inductor is in series with L . Therefore, the equivalent inductance is,

$$\begin{aligned} L_{eq} &= (2L \parallel 0.5L) + L \\ &= \left(\frac{2L \times 0.5L}{2L + 0.5L} \right) + L \\ &= \frac{L^2}{2.5L} + L \\ &= 0.4L + L \\ &= 1.4L \end{aligned}$$

Therefore, the equivalent inductance, L_{eq} is $\boxed{1.4L}$.

Step 4 of 7

(b)

Refer to Figure 6.77(b) in the textbook.

In the Figure 6.77(b), the two left side inductors are connected in parallel. The equivalent inductance is,

$$\begin{aligned} L_{eq1} &= (L) \parallel (L) \\ &= \frac{L \times L}{L + L} \\ &= \frac{L^2}{2L} \\ &= 0.5L \end{aligned}$$

In the Figure 6.77(b), the two right side inductors are connected in parallel. The equivalent inductance is,

$$\begin{aligned} L_{eq2} &= (L) \parallel (L) \\ &= \frac{L \times L}{L + L} \\ &= \frac{L^2}{2L} \\ &= 0.5L \end{aligned}$$

Step 5 of 7

Draw the reduced circuit of Figure 6.77(b).

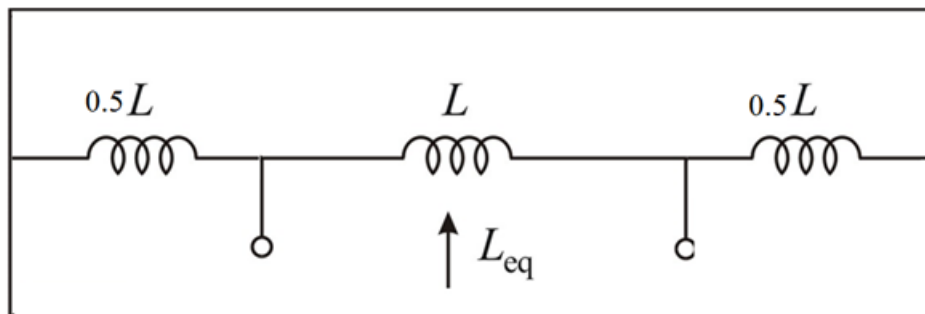


Figure 2

Step 6 of 7

Redraw the circuit in Figure 2.

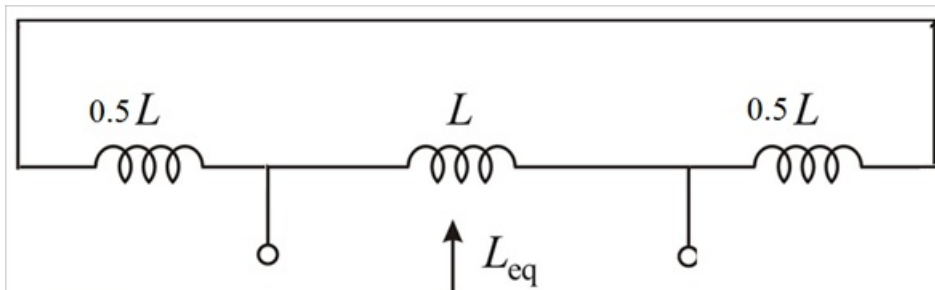


Figure 3

Step 7 of 7

In Figure 3, the inductors $0.5L$ and $0.5L$ are connected in series and the resultant is in parallel with L . Therefore, the equivalent inductance is,

$$\begin{aligned}L_{eq} &= (0.5L + 0.5L) \parallel (L) \\&= (L) \parallel (L) \\&= \frac{(L)(L)}{L + L} \\&= 0.5L\end{aligned}$$

Therefore, the equivalent inductance, L_{eq} is $\boxed{0.5L}$.